

期中考總複習

常見題型

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常見題型: Solving System
of Linear Equation

送分題

1. (20%) Let the 3×4 matrix A be defined by

$$A = [a_1 \ a_2 \ a_3 \ a_4] = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 1 & 1 \\ 1 & 2 & 1 & 0 \end{bmatrix}$$

(a) (5%) Find the reduced row echelon form of A . Denote it by R .

(b) (5%) Find the rank and nullity of A .

(c) (5%) Find the pivot columns of A .

(2013)

1. Let A and $\mathbf{d}$ be respectively defined by

$$A = \begin{bmatrix} 1 & -1 & 2 & 1 \\ -1 & 1 & 2 & 3 \\ 2 & -3 & 2 & 0 \\ -3 & 2 & 0 & 3 \end{bmatrix}, \quad \mathbf{d} = \begin{bmatrix} 2 \\ 6 \\ 3 \\ 9 \end{bmatrix}$$

(a) (10%) Find the reduced row echelon form of $[A \ \mathbf{d}]$.

(2014)

(b) (4%) Find the general solution to $A\mathbf{x} = \mathbf{d}$ in vector form.

(2018)

Solving System of Linear Equation

4. (15%) Let $A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3 \ \mathbf{a}_4 \ \mathbf{a}_5]$ be a 4×5 matrix and $\mathbf{b} \in \mathcal{R}^4$. The general solution to $A\mathbf{x} = \mathbf{b}$ is given by

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -5 \\ 0 \\ -3 \\ 2 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

(a) (3+2%) Find the rank and nullity of A .

(b) (3%) Find a basis for Null A .

(2015)

(c) (7%) Let $A' = [\mathbf{a}_2 \ \mathbf{a}_1 \ \mathbf{a}_3 \ \mathbf{a}_4 \ -\mathbf{a}_5]$ and $\mathbf{b}' = \mathbf{b} + \mathbf{a}_3$ Find the general solution to $A'\mathbf{x} = \mathbf{b}'$ in vector form.

4 x 6

反推 RREF

$$\text{RREF}([A \ \mathbf{b}]) = \begin{bmatrix} \\ \\ \\ \end{bmatrix}.$$

Solving System of Linear Equation

4. (15%) Let $A = [a_1 \ a_2 \ a_3 \ a_4 \ a_5]$ be a 4×5 matrix and $b \in \mathcal{R}^4$. The general solution to $Ax = b$ is given by

$$\begin{array}{lcl}
 & \begin{bmatrix} x_1 & & -5 & & -2 & & 1 \\ x_2 & & 0 & & 1 & & 0 \\ x_3 & = & -3 & + x_2 & 0 & + x_5 & 0 \\ x_4 & & 2 & & 0 & & -1 \\ x_5 & & 0 & & 0 & & 1 \end{bmatrix} & \begin{array}{l} \\ \\ \text{3 basic} \\ \\ \end{array}
 \end{array}$$

free ← (pointing to x_1)
free ← (pointing to x_5)

(a) (3+2%) Find the rank and nullity of A .

(b) (3%) Find a basis for Null A .

(2015)

(c) (7%) Let $A' = [a_2 \ a_1 \ a_3 \ a_4 \ -a_5]$ and $b' = b + a_3$ Find the general solution to $A'x = b'$ in vector form.

4 x 6

反推 RREF

$$\text{RREF}([A \ b]) = \left[\begin{array}{c} \\ \\ \\ \end{array} \right].$$

Solving System of Linear Equation

4. (15%) Let $A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3 \ \mathbf{a}_4 \ \mathbf{a}_5]$ be a 4×5 matrix and $\mathbf{b} \in \mathcal{R}^4$. The general solution to $A\mathbf{x} = \mathbf{b}$ is given by

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -5 \\ 0 \\ -3 \\ 2 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

(a) (3+2%) Find the rank and nullity of A .

(b) (3%) Find a basis for Null A . Solve $A\mathbf{x} = \mathbf{0}$

(2015)

(c) (7%) Let $A' = [\mathbf{a}_2 \ \mathbf{a}_1 \ \mathbf{a}_3 \ \mathbf{a}_4 \ -\mathbf{a}_5]$ and $\mathbf{b}' = \mathbf{b} + \mathbf{a}_3$ Find the general solution to $A'\mathbf{x} = \mathbf{b}'$ in vector form.

反推 RREF

$$\text{RREF}([A \ \mathbf{b}]) = \begin{bmatrix} 1 & 2 & 0 & 0 & -1 & -5 \\ 0 & 0 & 1 & 0 & 0 & -3 \\ 0 & 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Solving System of Linear Equation

4. (15%) Let $A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3 \ \mathbf{a}_4 \ \mathbf{a}_5]$ be a 4×5 matrix and $\mathbf{b} \in \mathcal{R}^4$. The general solution to $A\mathbf{x} = \mathbf{b}$ is given by

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -5 \\ 0 \\ -3 \\ 2 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

(a) (3+2%) Find the rank and nullity of A .

(b) (3%) Find a basis for Null A .

(2015)

(c) (7%) Let $A' = [\mathbf{a}_2 \ \mathbf{a}_1 \ \mathbf{a}_3 \ \mathbf{a}_4 \ -\mathbf{a}_5]$ and $\mathbf{b}' = \mathbf{b} + \mathbf{a}_3$ Find the general solution to $A'\mathbf{x} = \mathbf{b}'$ in vector form.

$$\begin{array}{ccc} \text{RREF}([A \ \mathbf{b}]) & \xrightarrow{\hspace{2cm}} & [A' \ \mathbf{b}'] \xrightarrow{\hspace{2cm}} \text{RREF}([A' \ \mathbf{b}']) \\ \begin{bmatrix} 1 & 2 & 0 & 0 & -1 & -5 \\ 0 & 0 & 1 & 0 & 0 & -3 \\ 0 & 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} & \begin{bmatrix} 2 & 1 & 0 & 0 & 1 & -5 \\ 0 & 0 & 1 & 0 & 0 & -2 \\ 0 & 0 & 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} & \begin{bmatrix} 1 & \frac{1}{2} & 0 & 0 & \frac{1}{2} & \frac{-5}{2} \\ 0 & 0 & 1 & 0 & 0 & -2 \\ 0 & 0 & 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

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常見題型: Determinant

Determinant

4. (6%) Let $A = \begin{bmatrix} 1 & 2c & 1-c \\ 0 & 3 & -1-c \\ 0 & -1-c & 3 \end{bmatrix}$. Calculate $\det A$ in terms of c . Determine the conditions on c so that A is invertible. (2018)

5. (5%) Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{bmatrix}$ and let $B = A - cI_3$. Calculate $\det B$ in terms of c . Determine the conditions on c so that B is invertible. (2012)

- How to compute $\det(A)$ (3x3 is usually good enough)
- A is invertible $\iff \det(A) \neq 0$

Determinant

1. (10%) Let A be an $n \times n$ matrix. Let R be the reduced row echelon form of A . Prove or disprove that $\det(A) = \det(R)$. (2015)

$$\det(AB) = \det(A) \det(B)$$

https://drive.google.com/file/d/1FCz1oB46ulzJx9ugdXiFJR_90J5ROnBM/view?usp=sharing (P8 – P11)

$$PA = R \quad P = E_k E_{k-1} \dots E_2 E_1 \quad E: \text{elementary matrix}$$

<https://drive.google.com/file/d/1TfKUor-7wfcwFtjrmFmbFlX9avBMeZp/view?usp=sharing> (P7)

$$\det(R) = \det(E_k) \det(E_{k-1}) \dots \det(E_2) \det(E_1) \det(A)$$

The answer is at

https://drive.google.com/file/d/1FCz1oB46ulzJx9ugdXiFJR_90J5ROnBM/view?usp=sharing (P4)

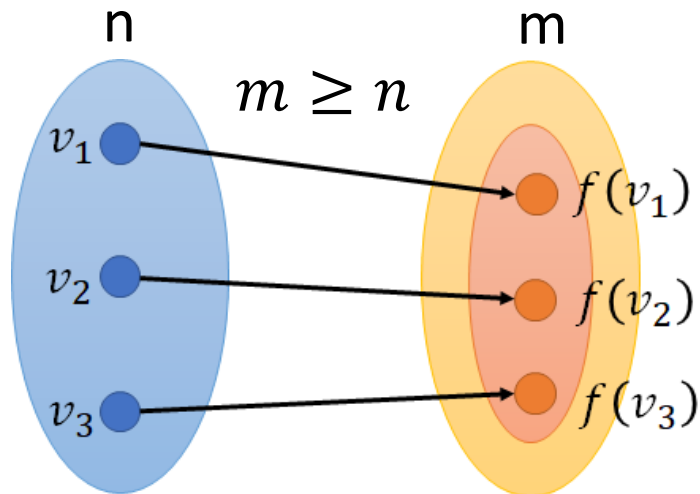
期中考總複習

常見題型: Linear Function

Linear Function

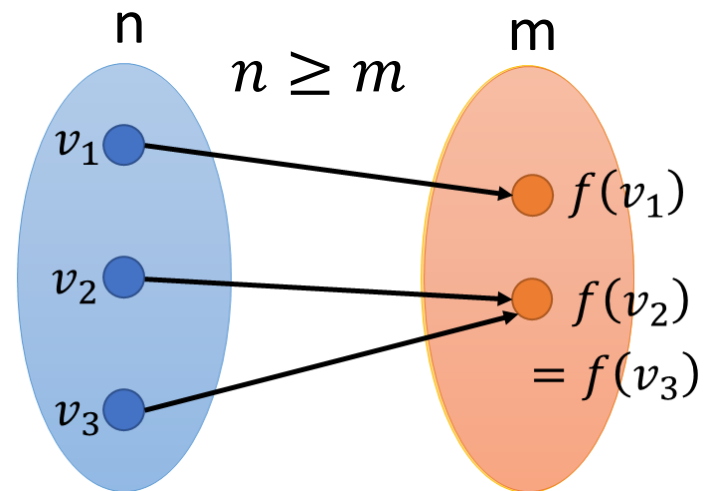
- Determining a function is linear or not.
- Determining a function is onto or one-to-one.

A: $m \times n$



one-to-one

The columns of A are independent.



onto

Rank A $\geq m$

Linear Function

4. In the following questions, T_1 , T_2 , and T_3 are linear transformations.

(a) (5%) Let $T_1 : \mathcal{R}^n \rightarrow \mathcal{R}^m$. Suppose $T_1(\mathbf{e}_i) = \mathbf{v}_i$, $i = 1, 2, \dots, n$ where $\mathbf{e}_i$ is the i th standard vector of $\mathcal{R}^n$. Find the standard matrix of T_1 . $[\mathbf{v}_1 \quad \cdots \quad \mathbf{v}_n]$

(b) (10%) Let $T_2 : \mathcal{R}^n \rightarrow \mathcal{R}^m$. Let $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ be a basis for $\mathcal{R}^n$ and suppose $T_2(\mathbf{u}_i) = \mathbf{w}_i$, $i = 1, 2, \dots, n$. Find the standard matrix of T_2 in terms of $\mathbf{u}_i$ and $\mathbf{w}_i$.

(c) (5%) Let $T_3 : \mathcal{R}^2 \rightarrow \mathcal{R}^3$. Suppose $T_3\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$ and $T_3\left(\begin{bmatrix} 2 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$. Find the standard matrix of T_3 .

(d) (5%) Is T_3 onto? Why?

(e) (5%) Is T_3 one-to-one? Why?

(2014)

$$A[\mathbf{u}_1 \quad \cdots \quad \mathbf{u}_n] = [\mathbf{w}_1 \quad \cdots \quad \mathbf{w}_n]$$

$$A = [\mathbf{w}_1 \quad \cdots \quad \mathbf{w}_n][\mathbf{u}_1 \quad \cdots \quad \mathbf{u}_n]^{-1}$$

5. (20%) Consider the following linear transformations: (2015)

$$T_1 : \mathcal{R}^2 \rightarrow \mathcal{R}^3 \text{ defined by } T_1 \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) = \begin{bmatrix} x_1 + x_2 \\ x_1 - 3x_2 \\ 4x_1 \end{bmatrix}$$

$$T_2 : \mathcal{R}^3 \rightarrow \mathcal{R}^2 \text{ defined by } T_2 \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = \begin{bmatrix} x_1 - x_2 + 4x_3 \\ x_1 + 3x_2 \end{bmatrix}$$

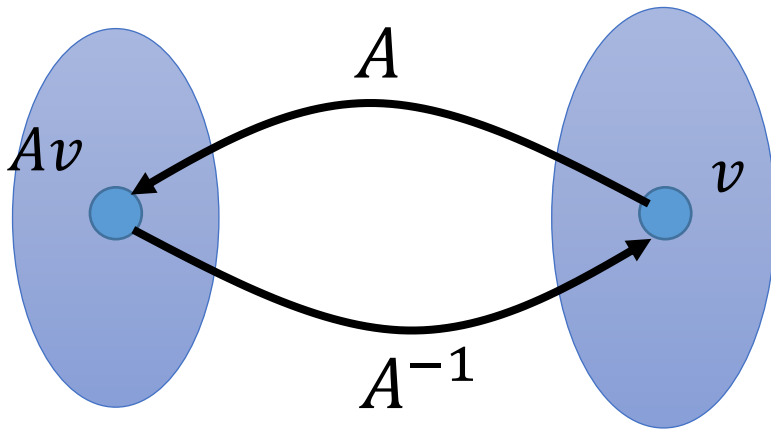
- (a) (8%) Find the standard matrices of T_1 , T_2 , T_1T_2 and T_2T_1 .
- (b) (4%) Is T_1T_2 onto? Is T_1T_2 one-to-one?
- (c) (4%) Let $U_1 : \mathcal{R}^n \rightarrow \mathcal{R}^m$ and $U_2 : \mathcal{R}^m \rightarrow \mathcal{R}^p$ be linear. Determine if the following statements are true or false (explain your answers):
 - (i) If $m > n$, then U_2U_1 cannot be onto.
 - (ii) If $m < p$, then U_2U_1 cannot be onto.
- (d) (4%) Let U_1 and U_2 be the linear transformations defined in (c). Prove that if U_1 and U_2 are one-to-one, then U_2U_1 is one-to-one.

期中考總複習

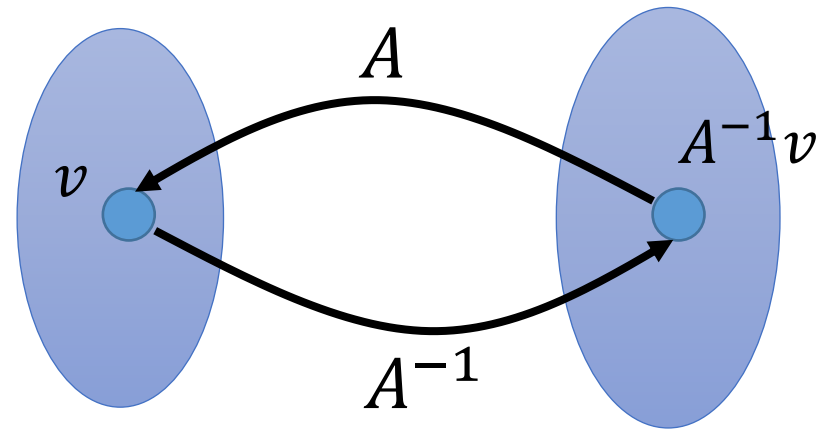
常見題型: Matrix Inverse

Invertible

- A is called invertible if there is a matrix B such that $AB = I$ and $BA = I$ ($B = A^{-1}$)



A must be one-to-one



A must be onto

(不然 A^{-1} 的 input 就會有限制)

Matrix Inverse

2. (10%) Find the inverse of the following matrix.

會不會算 A^{-1}

$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 8 \\ 2 & 5 & 6 \end{bmatrix} \quad (2018)$$

5. (15%) Let A be B be both invertible matrices of size $n \times n$. Answer whether the following statements are true or false. Justify your answer.

(a) (5%) $A + B$ is also invertible.

(b) (5%) AB is also invertible.

(c) (5%) $\det(B) \cdot A^{-1}$ is also invertible.

Inverse 的各種性質
(2016)

6. (15%) Consider the 2×4 matrix $A = \begin{bmatrix} 1 & 3 & 5 & 7 \\ 2 & 4 & 6 & 8 \end{bmatrix}$. (2016)

(a) (5%) Let R be the reduced row echelon form of A . Find R . (b) (5%) Find an invertible matrix P such that $PA = R$.

Inverse v.s. Row operation

Matrix Inverse

- 會不會算 A^{-1}
 - <https://drive.google.com/file/d/1DhsYYhM7jmJk9gE8OfY8m8KuVpeXZ74n/view?usp=sharing>
 - <https://youtu.be/YnvAlsJYBFg>
- Inverse 的各種性質
 - <https://drive.google.com/file/d/19xRjv261kE8WPwr2aAvdxaJiKbx9pLxd/view?usp=sharing>
 - <https://youtu.be/KnF2KvZa9tU>
- Inverse v.s. Row operation
 - <https://drive.google.com/file/d/1TfKUor-7wfcwFtjrmFmbFlX9avBMeZp/view?usp=sharing>
 - <https://youtu.be/BVyWqkkEUKA>

期中考總複習

常見題型: Subspace

Subspace

- 一定要會: 檢查一個 vector set 是不是 subspace

4. (10%) Let V and W be subspaces of R^n . Let $V + W = \{x + y : x \in V \text{ and } y \in W\}$. Prove that $V + W$ is a subspace of R^n . (2016)

注意:這不是聯集!

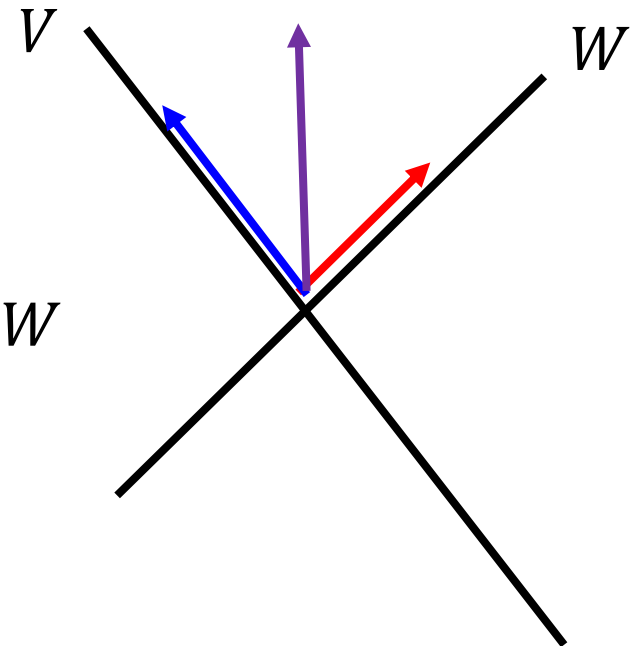
① $0 \in V, 0 \in W \Rightarrow 0 \in V + W$

② $x \in V + W \Rightarrow x = v + w$

$\Rightarrow cx = cv + cw \quad cv \in V, cw \in W$

$\Rightarrow cx \in V + W$

③ $x_1 \in V + W, x_2 \in V + W \dots\dots$



期中考總複習

常見題型: Dependent &
Independent

Dependent/Independent

3. (a) (6%) Let Q be an $n \times n$ invertible matrix. Let $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ be a linearly independent set of vectors in $\mathcal{R}^n$. Prove that $\{Q\mathbf{u}_1, Q\mathbf{u}_2, \dots, Q\mathbf{u}_k\}$ is linearly independent.

(2015)

$\{u_1, u_2, \dots, u_k\}$ independent



$\{Qu_1, Qu_2, \dots, Qu_k\}$ independent

$$c_1 Qu_1 + c_2 Qu_2 + \dots + c_k Qu_k = 0$$

$$\begin{array}{ccc} \text{blue arrow} & Q(c_1 u_1 + c_2 u_2 + \dots + c_k u_k) = 0 & \\ & Q^{-1} & Q^{-1} \end{array}$$

$$\begin{array}{ccc} \text{blue arrow} & c_1 u_1 + c_2 u_2 + \dots + c_k u_k = 0 & \text{blue arrow} \quad c_1 = c_2 = \dots = c_k = 0 \\ & \{u_1, u_2, \dots, u_k\} \text{ independent} & \end{array}$$

$\{u_1, u_2, \dots, u_k\}$ dependent

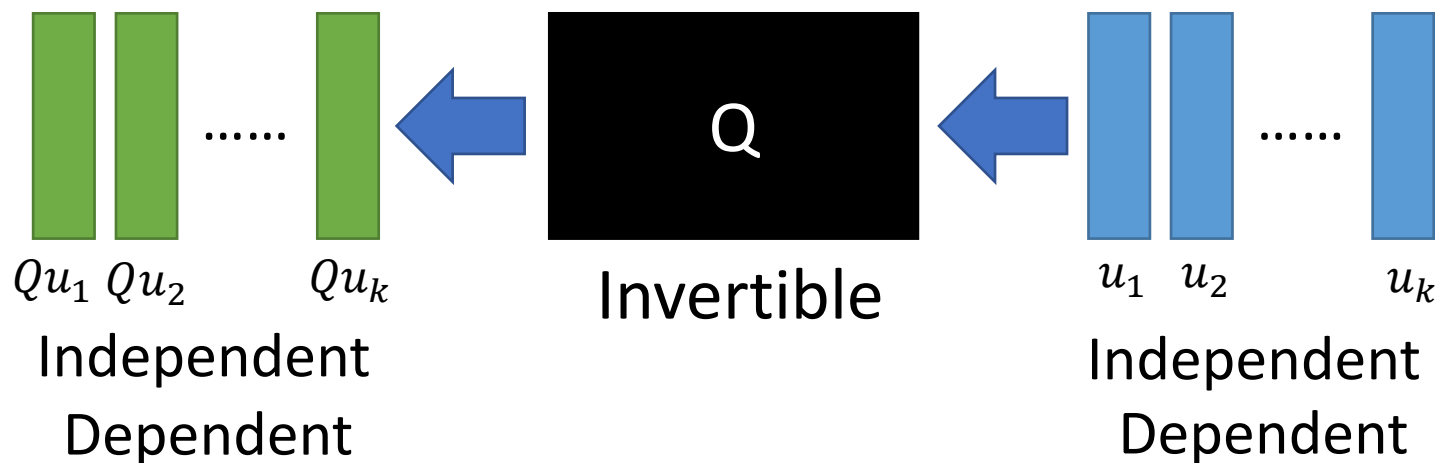


$\{Qu_1, Qu_2, \dots, Qu_k\}$ dependent

Dependent/Independent

3. (a) (6%) Let Q be an $n \times n$ invertible matrix. Let $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ be a linearly independent set of vectors in $\mathcal{R}^n$. Prove that $\{Q\mathbf{u}_1, Q\mathbf{u}_2, \dots, Q\mathbf{u}_k\}$ is linearly independent.

(2015)



7. (15%) Let A be an $m \times n$ matrix and R be its reduced row echelon form. We know that there is an $m \times m$ invertible P such that $PA = R$.

(2015)

- (a) (8%) Use the formula $PA = R$ to prove that the rows of R are linearly independent if and only if the rows of A are linearly independent.

期中考總複習

常見題型:

Rank v.s. Matrix Multiplication

Rank A (revisit)

Maximum number of Independent Columns

Number of Pivot Column

Number of Non-zero rows

Number of Basic Variables

Dim (Col A): dimension of column space = Dim (Row A)

Dimension of the range of A = Dim (Col A^T)

Full Rank: Rank = n & Rank = m

- The size of A is $m \times n$

$$\text{Rank } A = n$$

A is square or 高瘦

$$\begin{bmatrix} 1 & 0 & * & * & * & * \\ 0 & 1 & * & * & * & * \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & * & * & * & * \\ 0 & 0 & * & * & * & * \end{bmatrix}$$

$A\mathbf{x} = \mathbf{b}$ has at most one solution

The columns of A are linearly independent.

RREF of A: $\begin{bmatrix} I \\ 0 \end{bmatrix}$

All columns are pivot columns.

Full Rank: Rank = n & Rank = m

- The size of A is $m \times n$

$$\text{Rank } A = m$$

A is square or 矮胖

1	*	*	*	*	*
0	1	*	*	*	*
⋮	⋮	⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮	⋮	⋮
0	0	0	1	*	*
0	0	0	0	1	*

Every row of R contains a pivot position (leading entry).

$A\mathbf{x} = \mathbf{b}$ always have solution (at least one solution) for every $\mathbf{b}$ in $\mathcal{R}^m$.

The columns of A generate $\mathcal{R}^m$.

Rank

$$AB: m \times k$$

3. Let A and B be $m \times n$ and $n \times k$ matrices respectively.

(a) (4 + 2%) Prove that $\text{Col } AB$ is a subset of $\text{Col } A$. Also prove that $\text{rank } A \geq \text{rank } AB$.

(b) (4%) Suppose that $k = n$ and B is an invertible square matrix. Prove that $\text{Col } A$ is a subset of $\text{Col } AB$. $\text{Col } A = \text{Col } AB$ (2014)

2. (c) Suppose P is an $m \times k$ matrix and Q is a $k \times n$ matrix. Without using the inequality in (d) below, which is to be proved next, show that rank of the matrix $S = PQ$ is no greater than k . (10%)

Hint: Show first that columns of P form a generating set of $\text{Col } S$.

(d) Show that $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$ for any two matrices. (5%) **Hint:** Apply (a) to A and use (c). Then apply (a) to B and use (c) again. (2012)

Rank

https://drive.google.com/file/d/1brJ2BzNzxix2rozTOx7EAYY_AVJdto9g/view?usp=sharing (P4)

$$\begin{aligned}\text{Rank } A &= \text{Rank } A^T \\ \det A &= \det A^T\end{aligned}$$

3. Let A and B be $m \times n$ and $n \times k$ matrices respectively. (2014)

(a) (4 + 2%) Prove that $\text{Col } AB$ is a subset of $\text{Col } A$. Also prove that $\text{rank } A \geq \text{rank } AB$.

Columns of $AB \subset \text{Col } A \Rightarrow \text{Col } AB \subset \text{Col } A$

$\Rightarrow \dim(\text{Col } AB) \leq \dim(\text{Col } A) \Rightarrow \text{Rank } AB \leq \text{Rank } A$

(b) (4%) Suppose that $k = n$ and B is an invertible square matrix. Prove that $\text{Col } A$ is a subset of $\text{Col } AB$.

$$\text{Col } A = \text{Col } ABB^{-1} \subset \text{Col } AB \subset \text{Col } A$$

(d) Show that $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$ for any two matrices. (5%)

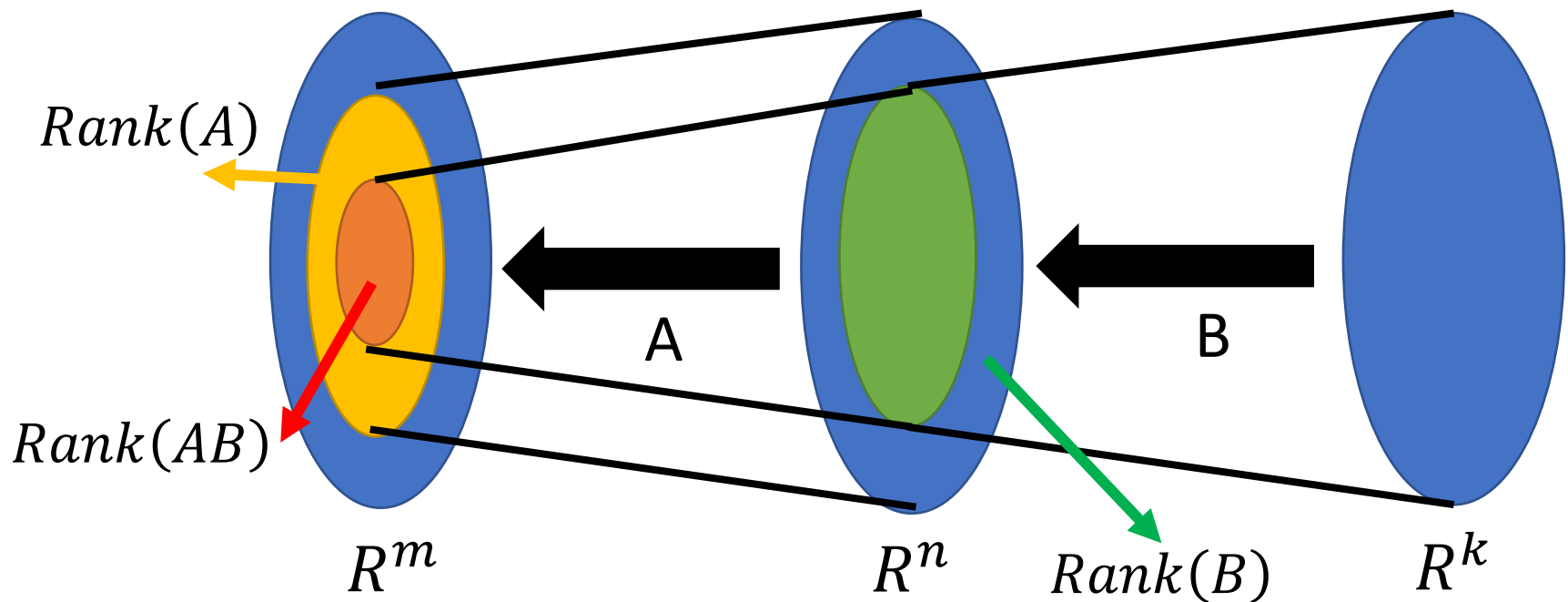
$$\text{Rank}(AB) = \text{Rank}(B^T A^T) \leq \text{Rank}(B^T) = \text{Rank}(B)$$

(b) (4%) Suppose that $k = n$ and B is an invertible square matrix. Prove that $\text{Col } A$ is a subset of $\text{Col } AB$.

Rank – Intuitive Explanation

- If A is a $m \times n$ matrix, and B is a $n \times k$ matrix.

$$\text{Rank}(AB) \leq \min(\text{Rank}(A), \text{Rank}(B))$$

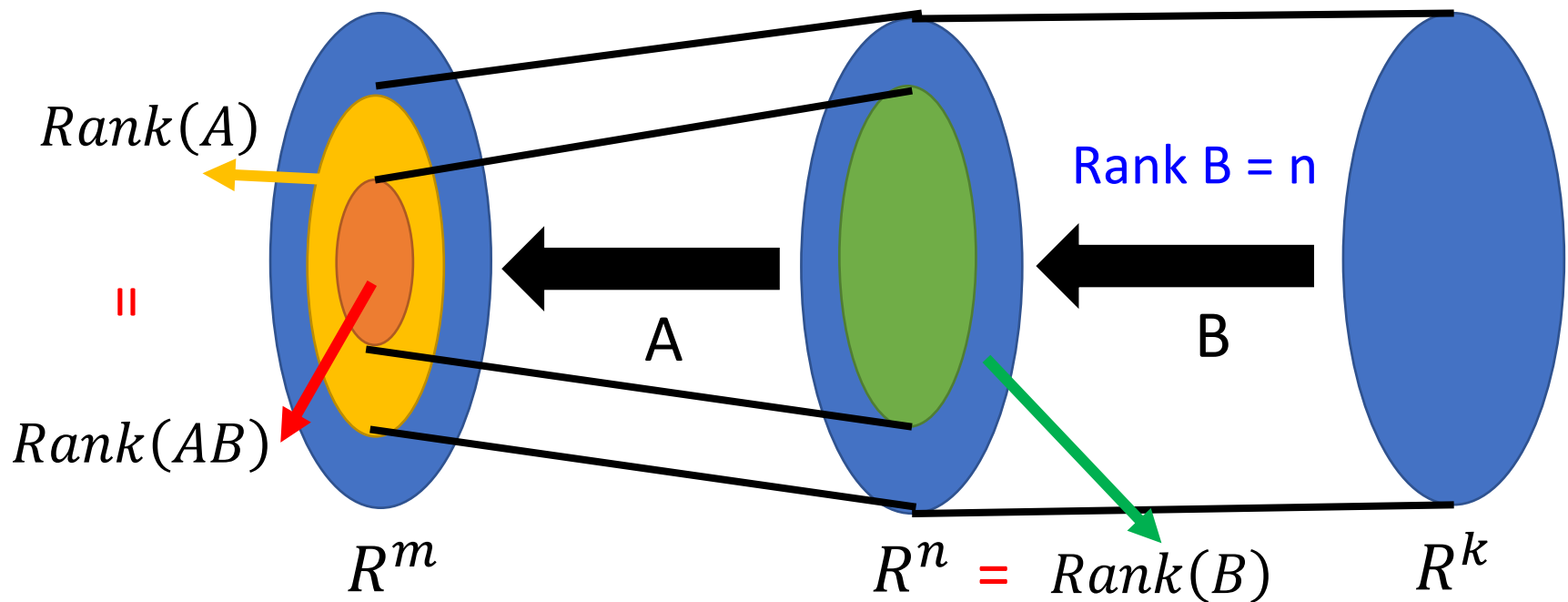


(b) (4%) Suppose that $k = n$ and B is an invertible square matrix. Prove that $\text{Col } A$ is a subset of $\text{Col } AB$.

Rank – Intuitive Explanation

- If A is a $m \times n$ matrix, and B is a $n \times k$ matrix.

$$\text{Rank}(AB) \leq \min(\text{Rank}(A), \text{Rank}(B))$$



期中考總複習

常見題型: Dimension & Basis

Review

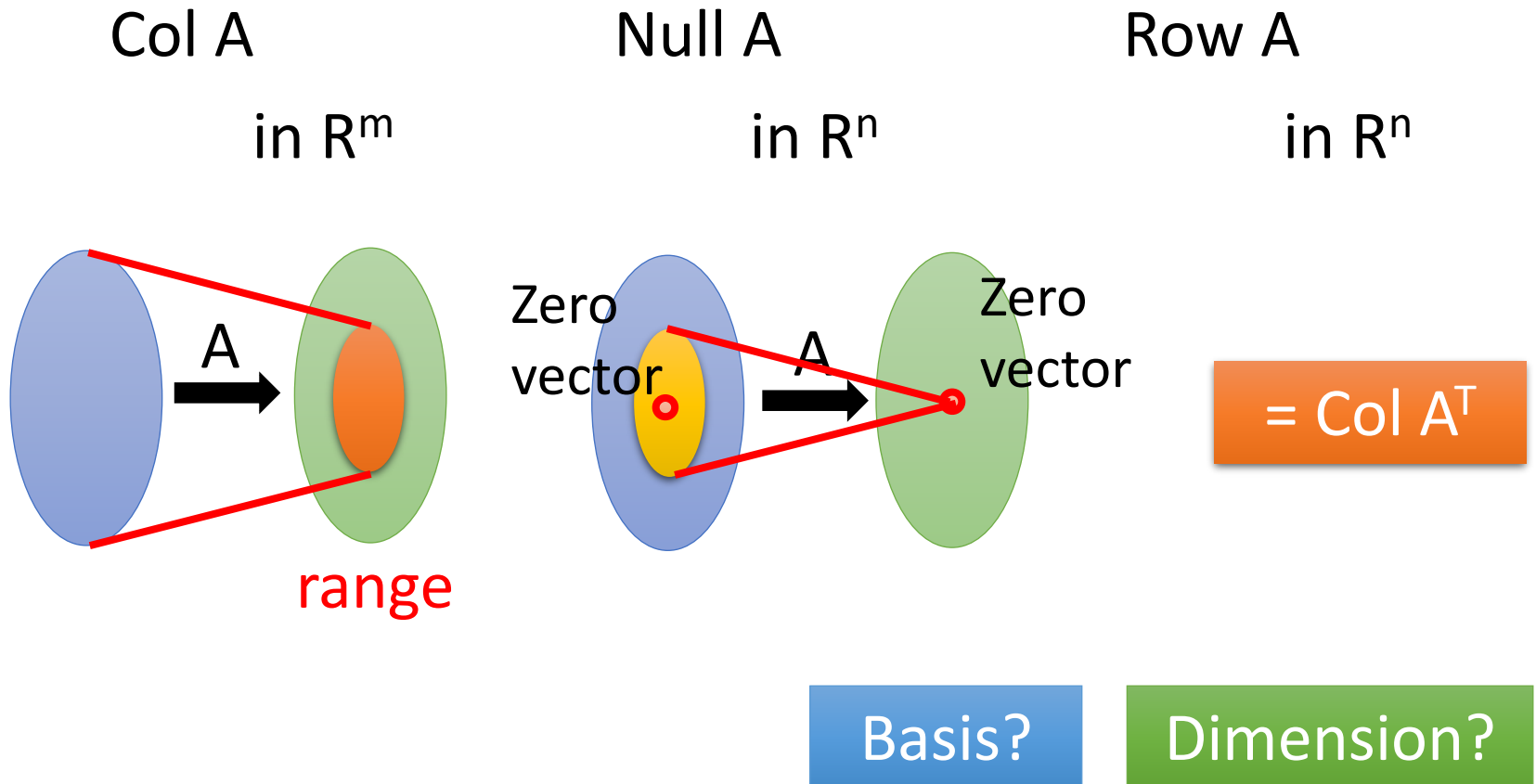
- Col A (range of A), $\dim(\text{Col A}) = \text{Rank}(A)$
 - Basis: Finding pivot columns of A
- Null A, $\dim(\text{Null A}) = \text{Nullity}(A)$
 - Basis: Solving $Ax = \mathbf{0}$
- Row A = Col A^T , $\dim(\text{Row A}) = \dim(\text{Col } A^T) = \text{Rank}(A^T) = \text{Rank A}$
 - Basis: Finding pivot columns of A^T

Original Matrix A v.s. its RREF R

	Relation	Span	Subspace
Column	Unchanged	Changed	$\text{Col A} \neq \text{Col R}$
Row	Changed	Unchanged	$\text{Row A} = \text{Row R}$

Three Associated Subspaces

- A is an $m \times n$ matrix



Summary

A is an $m \times n$ matrix

	Dimension	Basis
Col A	Rank A	The pivot columns of A
Null A	Nullity A $= n - \text{Rank A}$	The vectors in the parametric representation of the solution of $Ax=0$
Row A	Rank A	The nonzero rows of the RREF of A

Dimension & Basis

2. (10%) Let $T : \mathcal{R}^n \rightarrow \mathcal{R}^m$ be a linear transformation defined by $T(\mathbf{x}) = A\mathbf{x}$ for all $\mathbf{x} \in \mathcal{R}^n$, where A is an $m \times n$ matrix. Let $U : \mathcal{R}^m \rightarrow \mathcal{R}^n$ be a linear transformation defined by $U(\mathbf{x}) = A^T \mathbf{x}$ for all $\mathbf{x} \in \mathcal{R}^n$, where A^T is the transpose of A . Prove or disprove that the dimension of the range of T is the same as the dimension of the range of U . (2015)

$$\begin{aligned} T(x) = Ax &\Rightarrow \dim \text{ of Col } A = \text{rank of } A \\ U(x) = A^T x &\Rightarrow \dim \text{ of Col } A^T = \text{rank of } A^T \end{aligned} \quad \parallel$$

3. Answer whether the following statements are true or false. Justify your answer.
 (a) (5%) The mapping $\phi : R^n \rightarrow R^n$ defined by $\phi(\mathbf{x}) = [\mathbf{x}]_\alpha$ is a linear transformation, where $[\mathbf{x}]_\alpha$ is the coordinate vector of $\mathbf{x}$ relative to a basis α of R^n . (2016)

$$\begin{array}{ccc} & [x]_{\mathcal{B}} = B^{-1}x & \\ x & \xrightarrow{\hspace{10em}} & [x]_{\mathcal{B}} \\ & \xleftarrow{\hspace{10em}} & \\ & x = B[x]_{\mathcal{B}} & \end{array}$$

8. (15%) Let A be an $m \times n$ matrix, and let E_1 and E_2 be an $m \times m$ elementary matrix and an $n \times n$ elementary matrix, respectively. Determine the validity of each of the following statements. You don't have to give a proof if you determine a statement is true; however, if you think a statement is false, find a counterexample to show it (i.e., a specific example of matrices A and E_1 (or E_2).)

- (a) (3%) $\text{Col } A = \text{Col } E_1 A$. ☒
- (b) (3%) $\text{Row } A = \text{Row } E_1 A$. ☐
- (c) (3%) $\text{Col } A = \text{Col } AE_2$. ☐
- (d) (3%) $\text{Null } A = \text{Null } AE_2$. ☒
- (e) (3%) $\text{Null } A = \text{Null } E_1 A$. ☐

(2019)

$$\text{Row } A^T = \text{Row } E_2^T A^T$$